Cybersecurity Incident Report - Rayane Merlin

Section 1: Identify the type of attack that may have caused this network interruption
One potential explanation for the website's connection timeout error message is: The logs show that a lot of TCP SYN connection requests are sent by 203.0.113.0 to the server (192.0.2.1) in order to cause a DoS (Deny of Service). This event could be a TCP SYN Flood attack.

Section 2: Explain how the attack is causing the website to malfunction
When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake: 1. The client sends a SYN packet to request a connection 2. The server responds with a SYN-ACK packet to acknowledge 3. The client sends an ACK packet to validate the connection. Explain what happens when a malicious actor sends a large number of SYN packets all at once: The server receives a lot of connection requests and allocates resources to handle them correctly. This can exceed the server's resources and keep users from connecting. Explain what the logs indicate and how that affects the server: HTTP 504 Gateway Time-out errors appears to legitimate clients, indicating that users are experiencing connection problems due to the attack.